Bonus Assignment #1

Experiments:

Create, run, and test IPYNB scripts to run successfully the following **3 Experiments** for the provided requirements specification document of **Chicago WideCast Smart-Home Services**. You MUST create a NEW script and name it properly for every experiment:

Experiment Number	Platform	Model		Framework
1	Replicate [Cloud]	meta/meta-llama-3-70b-instruct	HuggingFaceEmbedding BAAI/bge-small-en-v1.6	LangChain/LangGraph
2	OpenAI [Cloud]	gpt-4o-mini	text-embedding-3-small	LangChain/LangGraph
3	Replicate [Cloud]	deepseek-ai/deepseek-r1	HuggingFaceEmbedding BAAI/bge-small-en-v1.6	LangChain/LangGraph

Assignment Deliverables:

You are required to submit on Canvas a SINGLE Zip file that has the following deliverables are:

1. Your IPYNB scripts
2. All of your source code and output
3. Analysis and Output report that has your assignment run, output, and analysis saved in Analysis.pdf
4. Video recording of 15-20 minutes as a demo for the run of your assignment using Panopto

Instructions:

1. **There is NO PARTIAL credit for the bonus assignment submission that has partial/incomplete code.**

2. **The grading for this assignment is BINARY: CREDIT or NO CREDIT**

3. Complete all the requirements listed in the **instructions** document.

4. All of your source code must be clearly documented and functional; **ZERO credit will be given to the submission that has nonfunctional code**.

5. Submit your comparative analysis report for the results you obtained for the experiments you executed.

6. Which model and platform produced the best results considering correctness, completeness, and coherence? Explain your answer.

7. **ZERO credit will be given to the submission that has NO analysis report.**

8. Submit your **IPYNB** scripts

9. **Panopto Video recording** (15-20 minutes) of your run that has your code and your output.